

Causal Directed Acyclic Graphs

A simple way to draw what you believe about cause and effect — and a rigorous way to read off exactly which variables an honest estimate must control for.

Based on the chapter by Ronja Foraita, Jacob Spallek & Hajo Zeeb · *Handbook of Epidemiology*

ROADMAP

From a Single Arrow to a Full Adjustment Strategy

- 1 **The visual language** — reading arrows, and the three basic patterns behind every diagram, through one running example.
- 2 **Blocking and opening paths** — the rule that decides whether a path carries a spurious association.
- 3 **Confounding and ground rules** — what confounding really means, and the rules a trustworthy diagram must follow.
- 4 **Finding the smallest fix** — a step-by-step method for finding the minimal set of variables that removes bias.
- 5 **When the full picture is missing** — six competing strategies for choosing adjustment variables without a complete diagram.
- 6 **Letting the data draw the diagram** — estimating a causal diagram directly from data, using computer algorithms.

What Is a Causal Diagram?

A causal diagram, or directed acyclic graph (DAG), is simply a picture: circles for variables, connected by arrows for the cause-effect relationships a researcher believes exist between them. “Directed” means every arrow points one way, from cause to effect. “Acyclic” means the arrows never loop back on themselves — nothing can end up causing itself.

The diagram is not a statistical result. It is a drawing of assumptions — a way of writing down, before any analysis is run, exactly what a researcher believes about how the variables in a study relate to one another.

Why Epidemiologists Draw Them

Every study that asks whether an exposure affects a disease faces the same hard question: which other variables must be statistically controlled for to get an unbiased answer? Controlling for too few leaves bias in. Controlling for too many, or the wrong ones, can create bias that was not there before.

A causal DAG, built from what is already known about a subject, turns that judgment call into a formal, checkable procedure — one that identifies exactly which variables belong in the adjustment set, instead of relying on rules of thumb like “adjust for everything available,” which can just as easily introduce bias as remove it.

BUILDING BLOCKS

Reading the Arrows

Every variable is a labeled box. An arrow from one box to another means the first is believed to directly influence the second. Two pairs of words describe these relationships, and they get used constantly from here on.



Parent and child — one direct step. X is a parent of M, and M is a parent of Y.

Ancestor and descendant — any number of steps. X is also an ancestor of Y, and Y a descendant of X.

BUILDING BLOCKS

Three Basic Patterns

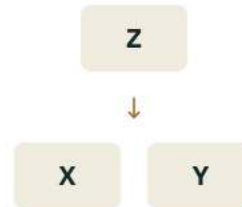
Any diagram, however large, is built from just three small local patterns. Recognizing them is the key to everything that follows.

Chain



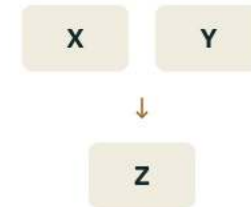
One thing causes the next, which causes the next.

Fork



One thing is a common cause of two others.

Collider



Two separate causes point to the same effect — also called a “v-structure.”

Every path in a larger diagram is just a sequence of these three (diagrams above are author-created illustrations). As the next few slides show, each behaves differently when you condition on it.

A RUNNING EXAMPLE

Meet the Breast Cancer Example

One hypothetical diagram runs through the rest of this guide. It asks whether having a first child at an older age affects a woman's risk of breast cancer, and it involves six variables.



Author-created illustration, not a reproduction of the original figure. SES = socio-economic status · FH = family history of breast cancer · BRCA = an inherited gene variant · E = education · D = first delivery at an older age, the exposure · BC = breast cancer, the outcome. The orange arrow is the relationship under investigation.

A RUNNING EXAMPLE

The Question Under Investigation

The arrow from D (first delivery at an older age) to BC (breast cancer) is deliberately left uncertain. Discovering whether it reflects a real effect, and how large, is the entire point of the analysis — but every other arrow in the diagram represents something a researcher already believes, based on prior knowledge, not something being tested.

This is exactly why the diagram matters before any statistics are run: family history, an inherited gene variant, education, and socio-economic status can all make the D–BC relationship look bigger, smaller, or entirely spurious in the raw data, depending on how they are handled. The rest of this guide uses this one diagram to work out precisely which of them must be accounted for.

BLOCKING AND OPENING PATHS

Blocking a Path

A “path” between two variables is any sequence of connected arrows joining them, regardless of which way the arrows point. If an open path exists, an association can flow along it — real or spurious. Whether a path is open or blocked depends on what happens at each variable along the way, and specifically on whether that variable has been **conditioned on**: held fixed, stratified by, or entered into an adjustment set.

The next two slides give the complete rule. Surprisingly, conditioning on a variable does not always block a path — for one of the three basic patterns, it does the opposite.

Chains and Forks: Easily Blocked

In both a chain and a fork, conditioning on the middle variable blocks the path between the two outer ones.

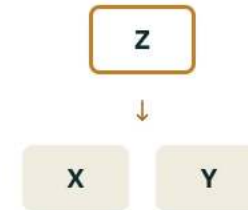
Chain: $X \rightarrow Z \rightarrow Y$



Z boxed — conditioned on

Once Z is held fixed, X carries no further information about Y through this path.

Fork: $X \leftarrow Z \rightarrow Y$



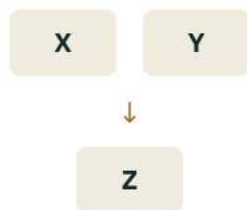
Z boxed — conditioned on

Once their common cause Z is held fixed, X and Y stop moving together through this path.

Colliders: Blocked by Default, Opened by Attention

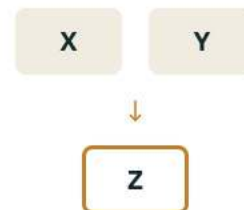
A collider works the opposite way, and this is the rule that trips people up most often.

Left alone



The path is already blocked. X and Y stay statistically unrelated through Z.

Conditioned on Z



The path opens. A new, artificial association appears between X and Y.

The same is true if you condition on anything downstream of Z. Conditioning on a collider — or on its effects — manufactures an association that was never really there.

PUTTING THE RULE TOGETHER

The D-Separation Rule

A path is **blocked** by a set of variables S if it contains either: a chain or fork variable that is inside S , or a collider variable where neither that variable nor anything it leads to is inside S .

Two variables (or groups of variables) are called **d-separated** by S if S blocks every single path between them. When that holds, no association — spurious or otherwise — can flow between them at all, given S .

This rule, formalized by the computer scientist Judea Pearl, is the single mathematical idea underneath everything a causal diagram is used for.

WHY IT MATTERS IN PRACTICE

Why Conditioning Can Backfire

Statisticians often draw a box around a variable's name to mark that it has been conditioned on. Two everyday habits turn out to be risky once you know the rule.

Adjusting for a mediator

A variable that sits directly on the causal path from exposure to outcome is a chain link, not a confounder. Conditioning on it blocks part of the very effect you are trying to measure, biasing the total effect toward looking smaller than it is.

Adjusting for a collider

A variable jointly caused by two other things looks, at first glance, like a perfectly reasonable thing to control for. Doing so opens a path that was safely closed, and manufactures bias where none previously existed.

The next section turns this rule into a precise definition of confounding, and the ground rules for drawing a diagram you can trust.

CONFOUNDING AND GROUND RULES

What Confounding Looks Like

A third variable confounds the relationship between an exposure and an outcome when it does two things at once: it independently causes the outcome, and it is also associated with the exposure. In diagram terms, this is exactly the fork pattern — a common cause sitting behind both, making them move together even if the exposure has no effect on the outcome at all.

A causal diagram lets you spot every such fork at a glance, and the d-separation rule tells you exactly which variables must be conditioned on to block it.

CONFOUNDING AND GROUND RULES

A Mediator Is Not a Confounder

A mediator sits on the causal chain between exposure and outcome — the exposure causes it, and it in turn helps cause the outcome. It looks, superficially, like exactly the kind of variable a careful analysis should control for. It is not.

The rule (Weinberg, 1993)

A variable that is partly caused by the exposure, and that changes the apparent exposure–outcome relationship, should not be adjusted for. Doing so removes part of the very causal effect the study is trying to estimate, not a source of bias.

CONFOUNDING AND GROUND RULES

Where the Assumptions Come From

A causal diagram does not discover anything by itself. Every arrow in it is a claim someone had to supply from outside the diagram — the diagram only works out the consequences of those claims.

Which variables to include, and which arrows to draw between them, is a subjective judgment — based on common sense, findings from other studies, and expert opinion in the relevant field. A diagram is only ever as trustworthy as the background knowledge used to build it.

CONFOUNDING AND GROUND RULES

Rules Every Diagram Must Follow

Every box is a variable

Not an event, and not a moment in time — a quantity that takes a value for each person in the study.

Every arrow is one claim

One arrowhead, one direct causal claim from one variable to another.

Unmeasured variables count too

If you believe an unmeasured factor matters, it still has to appear as its own box.

One limitation to hold onto from here on: the diagram shows only which variables are assumed to affect which others. It says nothing about how strong that effect is, or even whether it is protective or harmful — only whether it exists.

MINIMAL ADJUSTMENT SETS

The Back-Door Criterion

A **back-door path** between an exposure and an outcome is any path that starts by pointing backward into the exposure, rather than forward out of it — the route a spurious, non-causal association would travel.

A set of variables S satisfies the back-door criterion if:

1. S blocks every back-door path between the exposure and the outcome, and
2. Nothing in S was itself caused by the exposure.

Whenever such a set exists, adjusting for exactly those variables — and no others — yields an unbiased estimate of the exposure's causal effect. Statisticians call it a “sufficient adjustment set.”

MINIMAL ADJUSTMENT SETS

A Recipe for Finding Adjustment Sets

- 1 Draw the exposure and the outcome
- 2 Add every measured variable that plausibly matters
- 3 Add unmeasured variables you still believe exist
- 4 Mark every variable the exposure itself affects
- 5 List every back-door path between them
- 6 Cross off any path a collider already blocks
- 7 Find the smallest remaining set that blocks the rest
- 8 Check whether that choice opens a new path — if so, repeat from step 5

The next few slides walk through all eight steps on the breast cancer example.

STEPS 1–4 · APPLYING THE RECIPE

Building the Diagram, Step by Step

Steps 1–2

Start with D (first delivery at an older age) and BC (breast cancer). Add the measured variables that plausibly matter: E (education) and FH (family history).

Step 3

Add the unmeasured variables still believed to matter: SES (socio-economic status) and BRCA (an inherited gene variant), even though neither was recorded in the data.

Step 4

Mark what D itself affects. Here, only BC. That one arrow is set aside while the search below runs, since it is the effect under investigation, not a source of bias.



STEP 5

Finding Every Back-Door Path

Four paths run backward out of D and around to BC:

1 D ← E ← SES → FH ← BRCA → BC

2 D ← E ← SES → FH → BC

3 D ← FH ← BRCA → BC

4 D ← FH → BC

Every one of these could, in principle, make D and BC look related even if D has no real effect at all.

STEP 6

Erasing the Paths Colliders Already Block

Path 1 —
blocked



Here, FH sits between two arrowheads — $SES \rightarrow FH \leftarrow BRCA$ — a collider. Left alone, a collider already blocks its path. Path 1 needs no help; it disregards itself.

Three paths remain open, and every one of them passes through FH:



STEP 7

The Adjustment Set That Remains

All three remaining open paths share exactly one variable: FH, family history. Conditioning on FH alone blocks every one of them at once.

So far, {FH} looks like the minimal sufficient adjustment set — a single variable, doing the job of three separate paths.

But FH is not just an ordinary chain variable here. Recall it is also a collider — the meeting point of SES and BRCA. Conditioning on it is not free.

STEP 8

One Adjustment Can Trigger Another

FH is a collider between SES and BRCA. Conditioning on it blocks the three paths above — but, by the collider rule, it also opens a brand-new, purely statistical association between SES and BRCA that did not exist before.



A new, non-causal association — not a real arrow — sometimes called Berkson's paradox.

That new link opens yet another back-door path from D to BC, running through it. To close it too, the adjustment set needs one more member: either SES, or BRCA, or E. The valid candidates become **{FH, E}**, **{FH, SES}**, or **{FH, BRCA}**.

THE RESULT, AND ITS LIMIT

When the Answer Is Unmeasurable

Of the three valid candidates — {FH, E}, {FH, SES}, {FH, BRCA} — only one is actually usable. SES and BRCA were never measured in this study. The only feasible minimal sufficient adjustment set is **{FH, E}**.

The general lesson runs deeper than this one example: a diagram can only say **whether** a variable confounds, never **how strongly**. And sometimes, as here, every valid solution turns out to need a variable nobody recorded — in which case an unbiased estimate is simply out of reach by adjustment alone.

Later refinements (Shpitser et al., 2010; Perković et al., 2018) generalized this back-door logic into a graphical criterion proven to work in every case, not just this kind of example.

OTHER ADJUSTMENT APPROACHES

When the Whole Diagram Is Unknown

Everything so far assumed a complete, trustworthy diagram already in hand. In practice, researchers often know only part of the causal picture. Over the years, several rule-of-thumb strategies have been proposed for choosing adjustment variables under exactly that kind of incomplete knowledge.

The next slides walk through six such strategies. Each has genuine appeal, and each has a specific, well-documented way it can go wrong — a running comparison at the end applies all six to the same test case.

APPROACH 1 OF 6

Adjust for Everything Before Exposure

The idea: include every variable that is not itself a consequence of the exposure — anything measured, or determined, before the exposure occurred. It sounds comprehensive and safe.

Why it is not recommended

Elwert & Winship (2014) showed that bias can actually increase under this approach, if one of the pre-exposure variables happens to be a collider. “Before the exposure in time” is not the same guarantee as “safe to adjust for.”

APPROACH 2 OF 6

Adjust for Causes of Exposure

Pearl (2009) showed that the direct causes of the exposure form a mathematically sufficient adjustment set on their own.

Why it is not recommended

This set can include “instrumental variables” — things that cause the exposure but have no connection to the outcome except through it. Adjusting for them adds no protection against bias, but does inflate the statistical uncertainty of the estimate (Myers et al., 2011). Simulation studies by Witte & Didelez (2019) conclude this approach “cannot be recommended.”

APPROACH 3 OF 6

Adjust for Causes of the Outcome

The idea: include the direct causes of the outcome instead. Brookhart et al. (2006) and Witte & Didelez (2019) show this is often more statistically efficient — a more precise estimate — than several of the other approaches here.

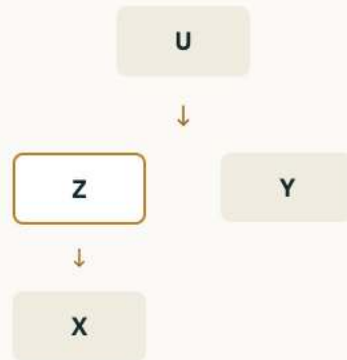
Why it can still fail

A direct cause of the outcome can also be a mediator — something the exposure itself partly causes on its way to affecting the outcome. Including one, as established earlier, biases the very effect being estimated.

APPROACH 4 OF 6

Adjust for Common Causes

The traditional textbook definition of a confounder: a variable that causes both the exposure and the outcome. Sufficient if every such variable is measured — but that is a demanding requirement, and the idea can mislead once an unmeasured variable enters the picture.



U, unmeasured, is the true common cause of X and Y. Z is not a common cause — it only leads to X. Yet adjusting for Z (boxed) blocks the same path just as well, because it is a chain link sitting along the way. A sufficient adjustment set need not literally contain “the” common cause (VanderWeele & Shpitser, 2011).

APPROACH 5 OF 6

Adjust for Any Plausible Cause

VanderWeele & Shpitser (2011) proposed a broader net: include every variable that is a cause of the exposure, **or** a cause of the outcome, or both. It works even with only partial knowledge of the diagram, as long as no unmeasured variable causes both.

The catch

Casting such a wide net tends to over-select variables, including exactly the instruments and mediators the earlier approaches struggled with.

A refined version

Excludes instruments and adds proxies for unmeasured common causes, reducing residual bias (VanderWeele, 2019) and praised for its simplicity (Ikram, 2019).

Witte & Didelez (2019) found it catches valid sets other approaches miss when many weak associations are present, though its larger sets can raise bias and variance when a study relies on matching.

APPROACH 6 OF 6

The Optimal Adjustment Set

This one asks a different question entirely: not just “which sets are valid,” but “which valid set gives the most precise estimate?” It requires knowing the full diagram, like the minimal sufficient set does — but optimizes for something extra.

The “O-set” (Henckel et al., 2022; Witte et al., 2020) includes every direct cause of the outcome, plus any variable lying on a causal path from exposure to outcome that helps rather than harms precision. It can end up larger than the bare minimal sufficient set — but that trade gives the smallest possible statistical uncertainty, in both ordinary regression and more flexible modeling.

PUTTING ALL SIX TO THE TEST

Comparing Six Approaches Side by Side



Also in the diagram: $Z3 \rightarrow Y$, and an unmeasured $U \rightarrow X$, $U \rightarrow Y$ — Z1 affects only X, not Y

Approach	What it selects here	Problem
Common cause	{U}	Unmeasured — unusable
Everything pre-exposure	{U, Z1, Z3, Z4}	Includes an instrument, plus unmeasured U
Causes of exposure	{U, Z1}	Includes an instrument, plus unmeasured U
Causes of outcome	{Z2, Z3, Z4}	Includes a mediator
Any plausible cause	{U, Z1, Z2, Z3, Z4}	Instrument, mediator, and unmeasured U, all at once
Optimal adjustment set	{Z3, Z4}	Valid, and most precise
Minimal sufficient set	{Z3}	Valid, and smallest

Only the two approaches that require knowing the full diagram come back clean. Every shortcut stumbles on this example in its own particular way.

CAUSAL DISCOVERY

Discovering a Diagram From Data

Everything so far started from a diagram a person drew, using existing knowledge. When reliable prior knowledge is scarce, contested, or simply doesn't exist yet, there is another option: let an algorithm estimate the diagram's structure directly from the patterns of association and independence already sitting inside the data.

This is called **causal discovery**, and it is a fundamentally more automated approach than anything covered so far.

CAUSAL DISCOVERY

Why More Than One Diagram Can Fit

Data alone cannot always tell diagrams apart. The chain, the reversed chain, and the fork below all predict the exact same statistical pattern — X and Y become independent once Z is held fixed — so no amount of data can distinguish which one is really true.

 $X \rightarrow Z \rightarrow Y$ $X \leftarrow Z \leftarrow Y$ $X \leftarrow Z \rightarrow Y$

These three are called “Markov equivalent.” A collider, $X \rightarrow Z \leftarrow Y$, is the exception — it implies a distinctive pattern all its own, so it can be identified uniquely.

Because of this, data-driven methods draw a **CPDAG** rather than a single DAG: undirected lines (X–Z) where the direction is genuinely ambiguous from data, and arrows only where the direction is uniquely determined — as at a collider.

CAUSAL DISCOVERY

Too Many Diagrams to Search by Hand

Why not just try every possible diagram and see which fits best? Because the number of possible diagrams explodes far faster than intuition suggests, even for a modest number of variables.

With the six variables of the breast cancer example

3,781,503

possible diagrams

With just ten variables

$\approx 4.8 \times 10^{18}$

possible diagrams

Checking every candidate one by one is hopeless. Every practical method instead uses a smart search strategy.

CAUSAL DISCOVERY

Two Search Strategies

Score-based

Give every candidate diagram a numeric score for how well it fits the data, and search for the best-scoring one without checking all of them.

Constraint-based

Run a sequence of statistical independence tests, and build the diagram's edges one at a time from the results.

Both are estimates, built from a finite, noisy sample — not a guaranteed recovery of the one true diagram.

CAUSAL DISCOVERY

Score-Based Search

Each candidate diagram receives a score that rewards fitting the data well and penalizes needless complexity, so the search doesn't simply favor the most tangled diagram available.

Since checking every diagram is impossible, most methods search **greedily** — taking whichever small step improves the score next, and stopping once no local improvement remains. This finds a good diagram, not provably the best one overall.

Greedy Equivalence Search (GES, Chickering, 2002) is the prototypical method; it searches directly over CPDAGs, since Markov-equivalent diagrams share the same score anyway. GIES (Hauser & Bühlmann, 2012) extends the idea to data that includes deliberate interventions.

Most score-based methods assume there is no unmeasured common cause anywhere in the picture — a strong assumption, and often an unrealistic one in epidemiology.

CAUSAL DISCOVERY

Constraint-Based Search

This family works the other way around: start by assuming every variable might be connected to every other, then use statistical independence tests, one pair at a time, to decide which connections the data actually supports. An edge is removed whenever a pair tests as independent. What is left is the diagram's "skeleton," which is then partly oriented into a CPDAG.

The right independence test depends on the data: chi-squared tests for categorical variables, partial-correlation tests for roughly continuous ones, and newer flexible tests for relationships that are neither.

Choosing how strict each test should be is itself debated — some researchers control the rate of false discoveries directly, others treat strictness as a setting to be tuned against the data.

CAUSAL DISCOVERY

The PC Algorithm, in Three Steps

The most prominent constraint-based method, named for its creators Peter Spirtes and Clark Glymour. A later refinement, PC-stable, ensures the result no longer depends on the order variables happen to be listed in.

- 1 Start fully connected. Test every pair of variables for independence, and remove the edge between any pair found independent, leaving a “skeleton.”
- 2 For every remaining $X-Z-Y$ trio, check what made X and Y independent. If Z was not needed for that, orient it as a collider, $X \rightarrow Z \leftarrow Y$.
- 3 Apply a small set of logical rules to orient any further edges possible, without creating a new cycle or an unjustified collider.

CAUSAL DISCOVERY

Watching the PC Algorithm Work

Run on data simulated from the breast cancer diagram, without telling the algorithm the diagram's true shape:

Step 1 — building the skeleton

SES—BRCA is dropped: the two test as independent overall. SES—D is also dropped: they test as independent once E is held fixed.

Step 2 — spotting a collider

SES and BRCA are independent on their own, but not once FH is added to the picture — exactly the signature of a collider. $SES \rightarrow FH \leftarrow BRCA$ gets oriented with confidence.

Step 3 — what's left undirected

Most remaining edges resolve to match the original diagram, but a few — including some within the SES, E, and D trio — stay undirected. The data alone cannot break every tie.

CAUSAL DISCOVERY

A Cousin Algorithm for Hidden Variables

The PC algorithm assumes there is no unmeasured common cause anywhere — often unrealistic in epidemiology. FCI (Fast Causal Inference), from the same research group, relaxes exactly that assumption, and stays asymptotically correct even when hidden confounding is present.

The trade-off

FCI's output is a “Partial Ancestral Graph” (PAG), which typically carries less definite causal-direction information than a CPDAG. PAGs use extra ambiguous edge marks, such as circles, to flag exactly where hidden confounding might explain the data — more honest about the added uncertainty, but correspondingly harder to read.

PUTTING IT INTO PRACTICE

Software That Does This for You

dagitty.net — the most widely used tool for drawing a diagram and reading off adjustment sets, free and browser-based; also available as the R packages *dagitty* and *ggdag*.

pcalg (R) — finds adjustment sets, including the optimal one, and implements the PC, FCI, and score-based search algorithms.

micd and **tpc** (R) — causal discovery adapted for missing data and mixed measurement scales, or for data with a known time order.

bnlearn (R and Python) — a broad library of causal discovery algorithms, plus tools for learning parameters and running inference.

TETRAD — a standalone desktop application bundling many search algorithms, also usable from R or Python.

All of these still require judgment to use well — the software finds structure in data, but interpreting it responsibly remains a human task.

HONEST LIMITATIONS

What a Diagram Cannot Tell You

No interactions

A standard diagram cannot show that one variable's effect depends on the level of another. Proposals to add this exist, but only under strict assumptions, and it remains a genuinely open problem.

No size, no direction

An arrow says nothing about whether an effect is protective or harmful, nor how large it is. A diagram can flag that bias is likely; it cannot say how much.

Both limits are worth remembering every time a tidy diagram makes a messy causal question look more settled than it really is.

HONEST LIMITATIONS

One Diagram Is Rarely the Whole Story

A hand-drawn diagram is built on a single hypothesis, and most of the background knowledge that goes into it is itself based on plain, non-causal associations from earlier studies. No diagram built this way can be fully guaranteed correct.

A better habit than trusting one diagram

Shrier & Platt (2008) recommend drawing several alternative, equally plausible diagrams, checking whether they lead to the same adjustment set, and publishing this comparison alongside the main analysis (Geneletti et al., 2011).

Data-driven diagrams have their own version of this problem: a single estimated graph is a best guess from one sample, Markov-equivalent alternatives can tell very different causal stories from identical data, and real-world causal discovery is still more common in genetics than in epidemiology.

BOTTOM LINE

A Rigorous Way to State What You Believe

A causal diagram turns an informal, easy-to-get-wrong judgment call — what should I adjust for? — into an explicit, checkable procedure. The back-door criterion tells you exactly which variables belong in that set, and exactly why.

It cannot replace domain expertise, cannot quantify an effect's size on its own, and is only ever as sound as the assumptions drawn into it. But making those assumptions explicit, and open to challenge, is precisely the point.

SOURCE & ATTRIBUTION

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